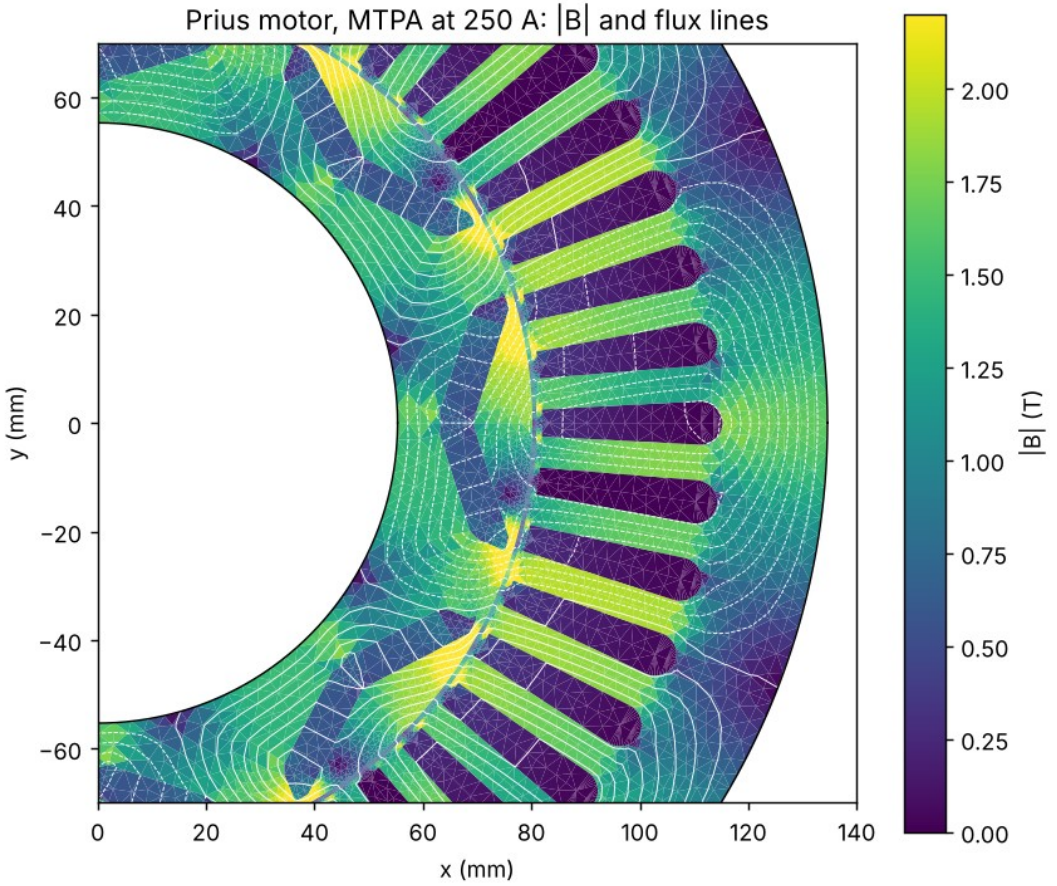
2004 Toyota Prius traction motor: verified FEA to drive, thermal and design

Open-source chain in PINNeAPPle (pinneapple_simulation.electromagnetics.machines):

2-D nonlinear FEA -> second-code cross-check -> drive (MTPA, flux weakening, WLTC) -> end effects -> thermal and demagnetisation -> skew and tolerances -> FEA + GP + NSGA-II design -> exchange formats

Every number in these slides is a measured output of examples 10 to 17 (results/*.json), regenerated by this script.

Geometry and materials: pyleecan's Toyota_Prius.json (48 slots, 8 poles, V magnets, M400-50A).



Limits: a model study from public data; no test-bench measurement of this motor was available.

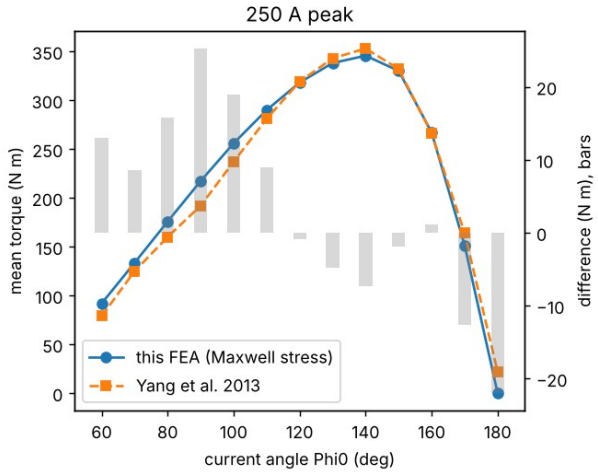
2-D FEA, verified

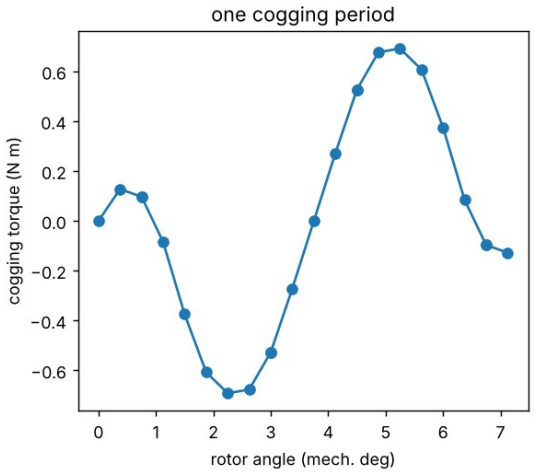
Nonlinear A_z FEM, conforming mirrored meshes, Arkkio torque: 29936 elements.

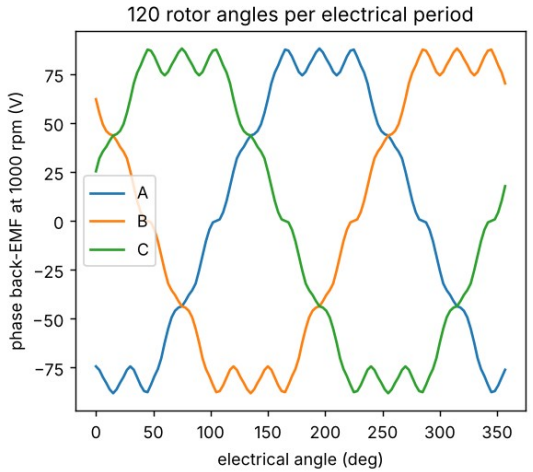
Verification cases against closed forms: 6 of 6 pass (magnetised ring: second order, as P1 should).

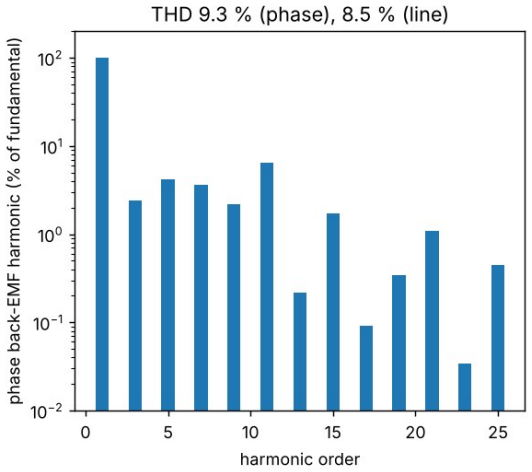
Torque vs current angle at 250 A against Yang et al. 2013: peak 345.6 vs 353 N m, rms difference 13.3 N m (their model and the pyleecan geometry differ: centre post, steel data).

Cogging 1.38 N m pk-pk; back-EMF fundamental 87.6 V peak at 1000 rpm.









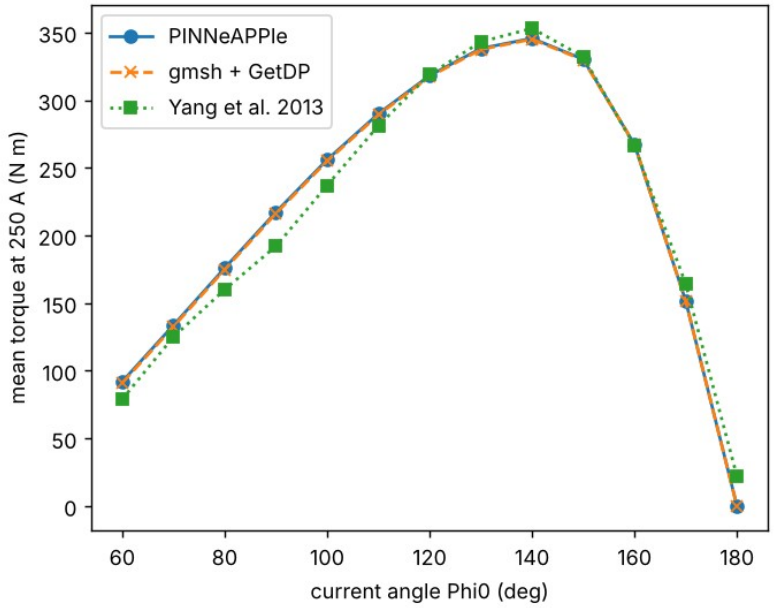
Limits: 2-D only; P1 elements; iron loss from B waveforms without minor loops.

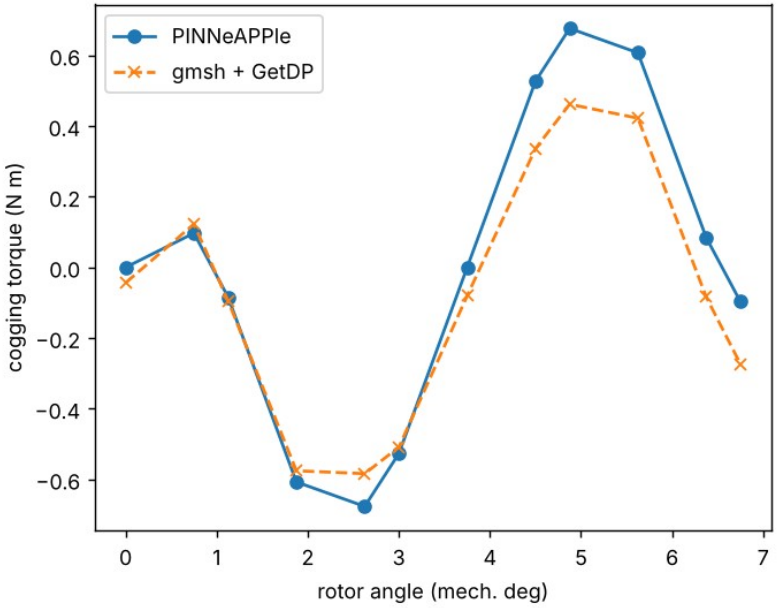
Second code: gmsh + GetDP

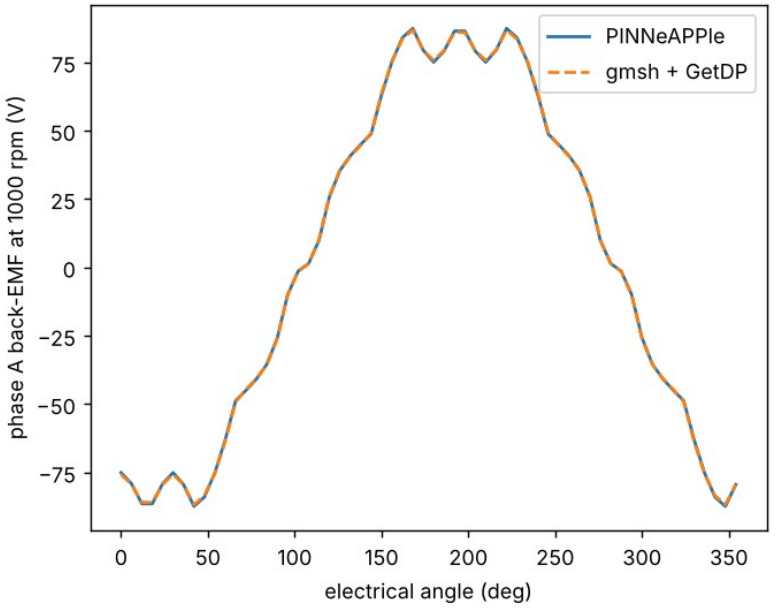
Same geometry and materials, GetDP's own mesh, Newton solver and torque: torque vs angle within 1.09 N m.

Back-EMF fundamental 87.60 vs 87.64 V; L_d 1.370 vs 1.385 mH; L_q 1.856 vs 1.864 mH.

Cogging does not agree: 1.34 vs 1.00 N m at 1920 gap nodes, both converged; with linear iron 1.223 vs 1.211 N m -> the difference is in the saturated bridges.







Limits: cogging uncertain by about 25 % between two codes; FEMM, pyleecan, Elmer not run (inputs written).

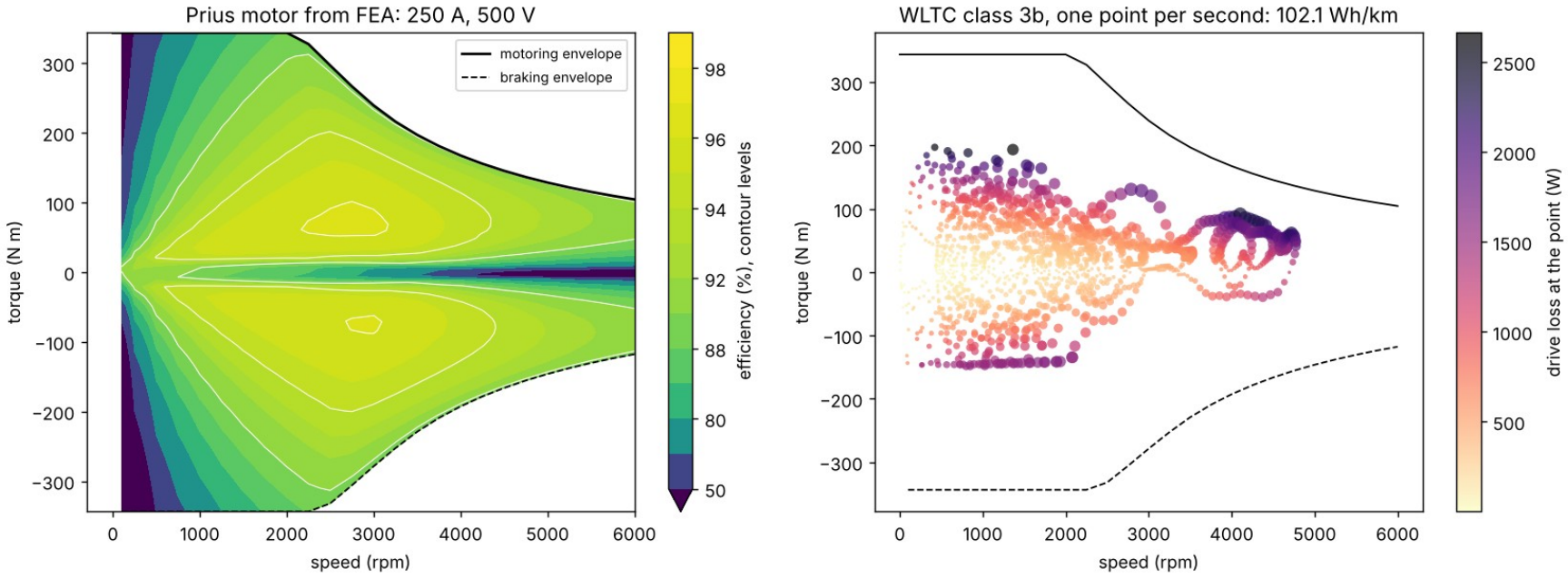
Drive: MTPA, flux weakening, WLTC

FEA flux map (saturation, cross-saturation), 250 A, 500 V DC with SVPWM: 343.6 N m at low speed, 104.4 N m at 6000 rpm, 77.7 kW peak.

Peak efficiency 96.2 % at 2750 rpm, 70.7 N m (copper + iron + magnet loss).

WLTC class 3b, illustrative compact car: 102.1 Wh/km at the battery (drive losses only).

Optimiser checked against the closed forms of a linear machine (1e-14) and energy bookkeeping.



Left: efficiency = P / (P + losses) motoring, (|P| - losses) / |P| braking; contour levels 50, 70, 80, 85, 88, 90, 92, 93, 94, 95, 96, 97, 98, 99 %; white lines 90, 94, 96 %. Torque grid: fractions of the envelope at each speed (100 to 6000 rpm); zero torque (undefined efficiency) is the gap between the halves. Right: colour = copper + iron + magnet loss, marker area proportional to the energy |P| dt of the second.

Limits: steady state; no inverter, mechanical or battery losses; the car is not a measured Prius.

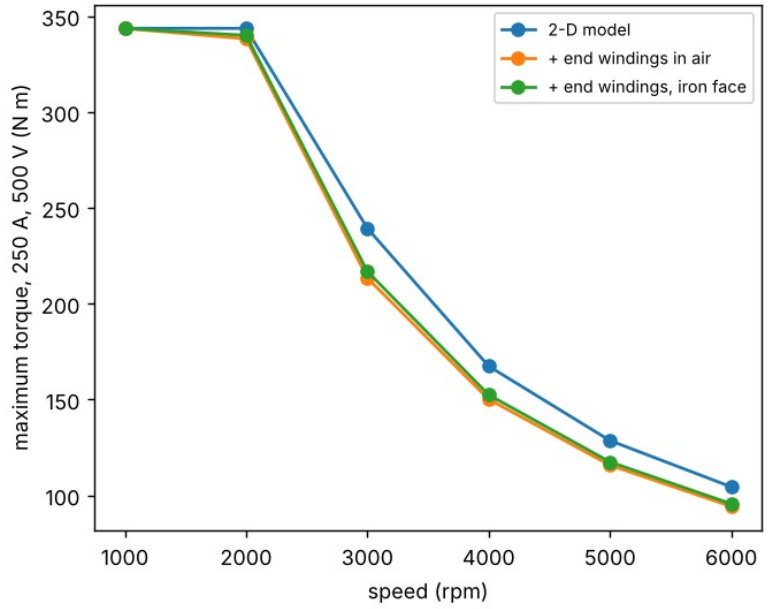
End effects

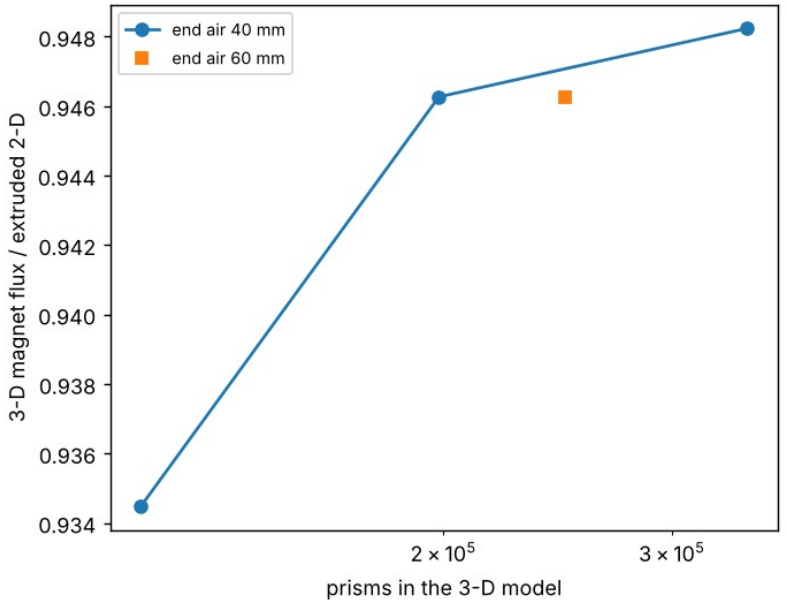
End windings (Neumann, thin filaments, verified against Rosa and Maxwell): L_end 135.2 uH in air, 115.3 uH with the iron face.

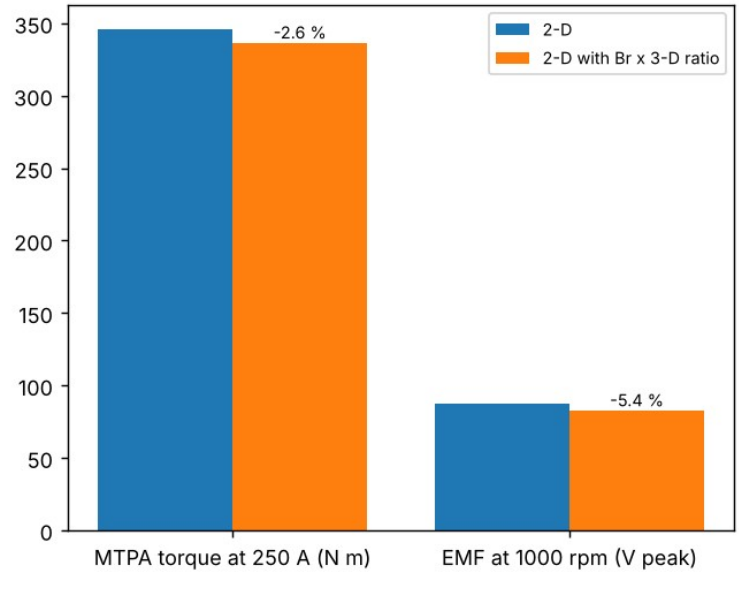
Effect on the envelope at 3000 rpm: 239.3 -> 213.3 N m.

Magnet end leakage, 3-D GetDP model of half the stack: flux ratio 0.9345 -> 0.9463 -> 0.9482 on three meshes.

Corrected 2-D model: MTPA torque -2.6 %, back-EMF -5.4 %.







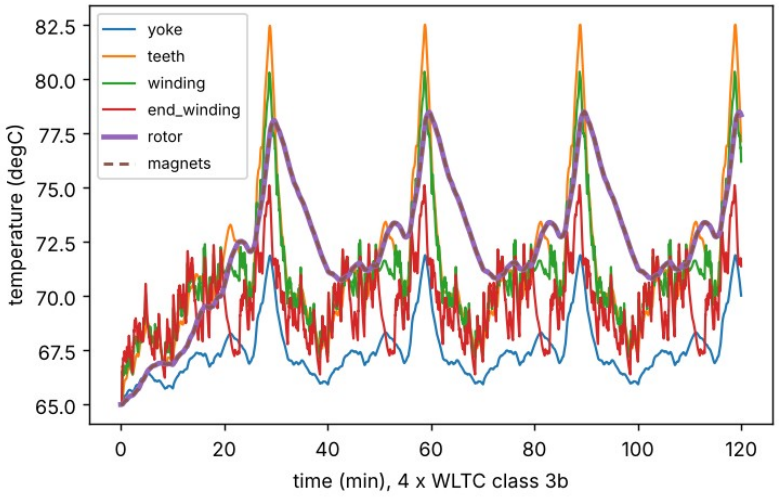
Limits: corrections, not a coupled 3-D solution at load; no measured end-winding inductance to compare.

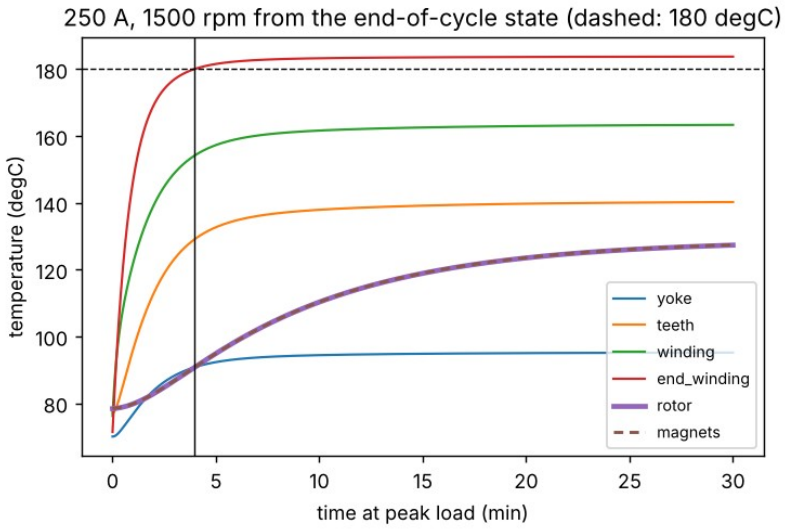
Magnet temperature and thermal network

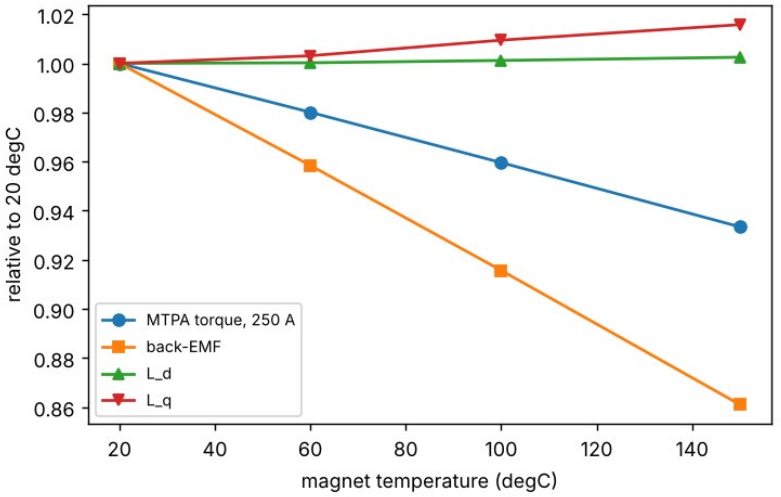
Magnets 20 -> 150 degC (Br -0.1 %/K): torque at 250 A 345.6 -> 322.6 N m, back-EMF 87.6 -> 75.4 V.

4 x WLTC from the drive losses: teeth 82.5, winding 80.3, magnets 78.5 degC.

Peak load from that state: hotter winding node at 180 degC after 237 s; magnets settle at 127 degC.







All nodes start at the coolant temperature 65 degC. Boundaries: water jacket on the stator back (h A 300 W/K, coolant 65 degC), end windings to 65 degC air (h 40 W/m^2 K), rotor through shaft and bearings (2 W/K to 65 degC). Rotor and magnet curves coincide (dashed magnets over the rotor): the magnet loss is about 0.2 W, the rotor-magnet contact about 340 W/K. Thermal data are illustrative defaults (MachineThermalData), not measured on a Prius.

Limits: seven-node network with illustrative thermal data (not fitted to a Prius); one temperature per node.

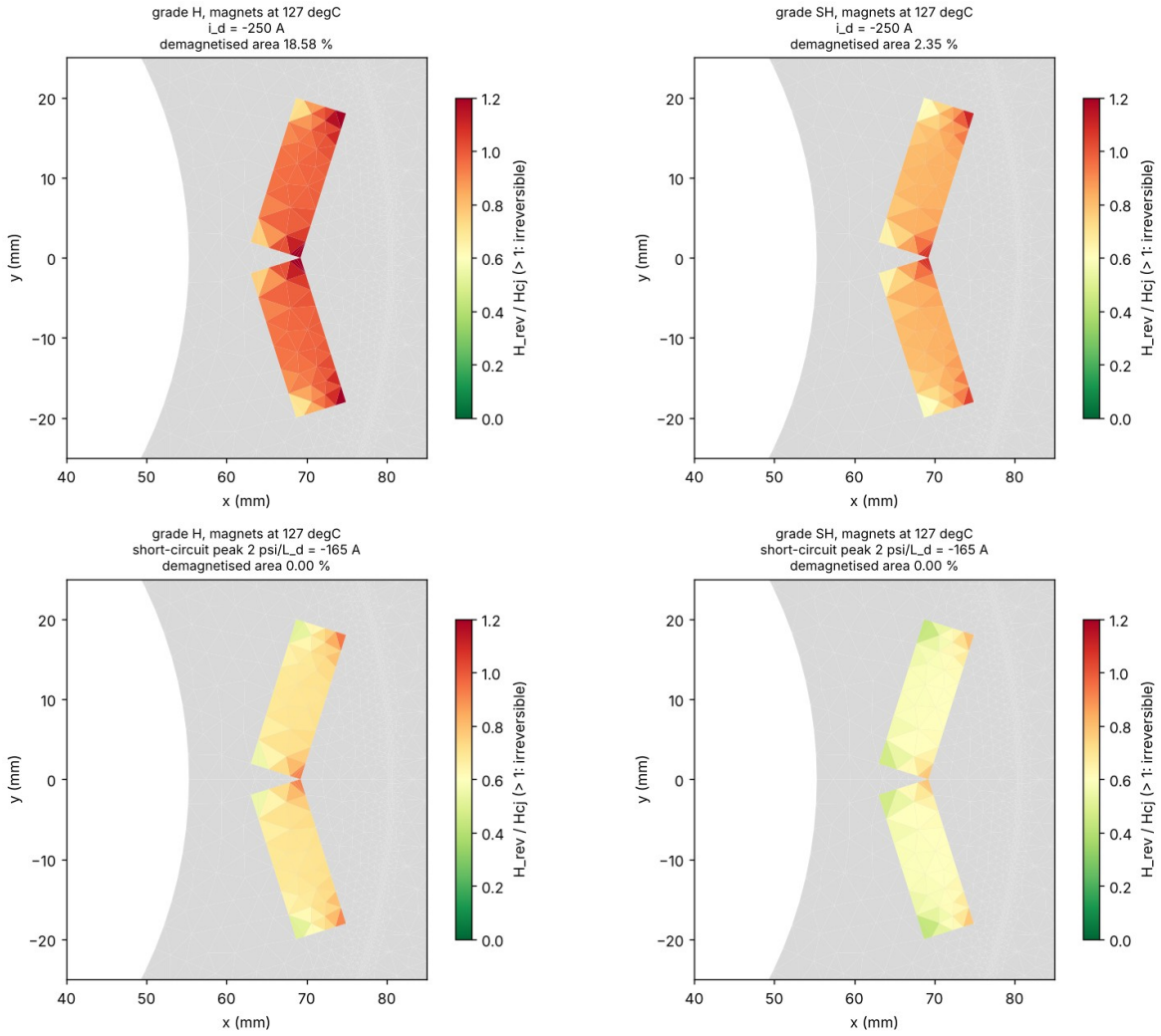
Demagnetisation

Reverse field per magnet element against Hcj(T) of the grade letters, magnets at 127 degC.

Full d-axis current (-250 A): demagnetised area N 99.4 %, M 93.3 %, H 18.6 %, SH 2.4 %, UH 0.0 %.

Short-circuit estimate: N 31.8 %, M 3.2 %, H 0.0 %, SH 0.0 %, UH 0.0 %.

Mesh study inside the magnets (4 levels): grade H at -250 A 18.6 to 19.4 % -> area fractions converge; the worst element does not (sharp corners).



Limits: Hcj coefficient assumed (-0.55 %/K); static knee test; the Prius magnet grade is not public, the rows compare candidate grades.

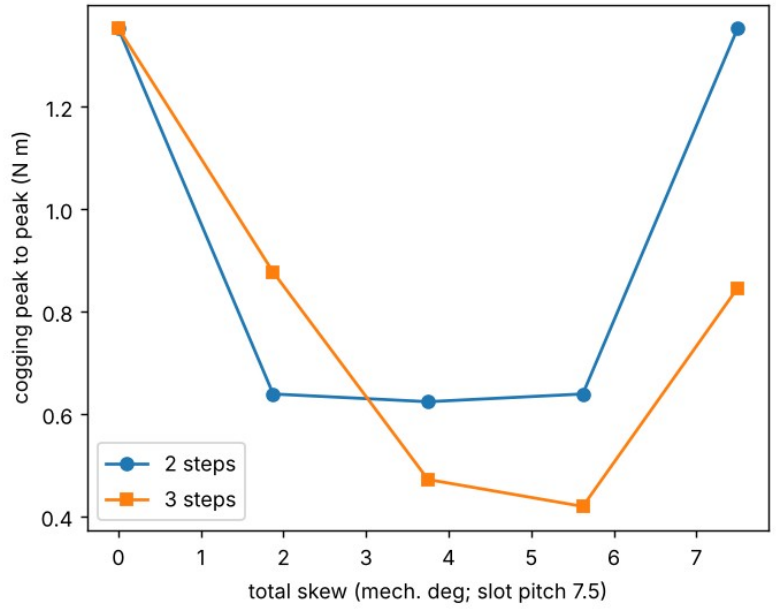
Step skew

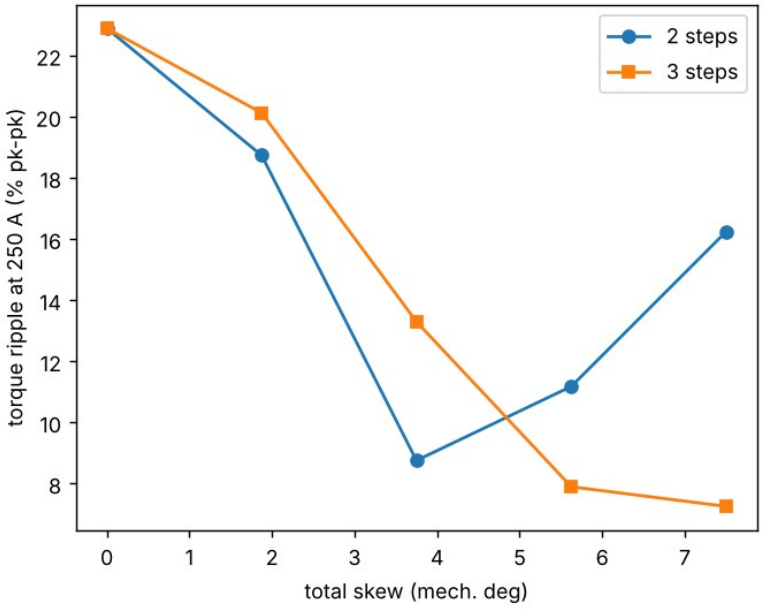
Multi-slice FEA. Straight rotor: cogging 1.35 N m, ripple 22.9 %, torque 345.6 N m.

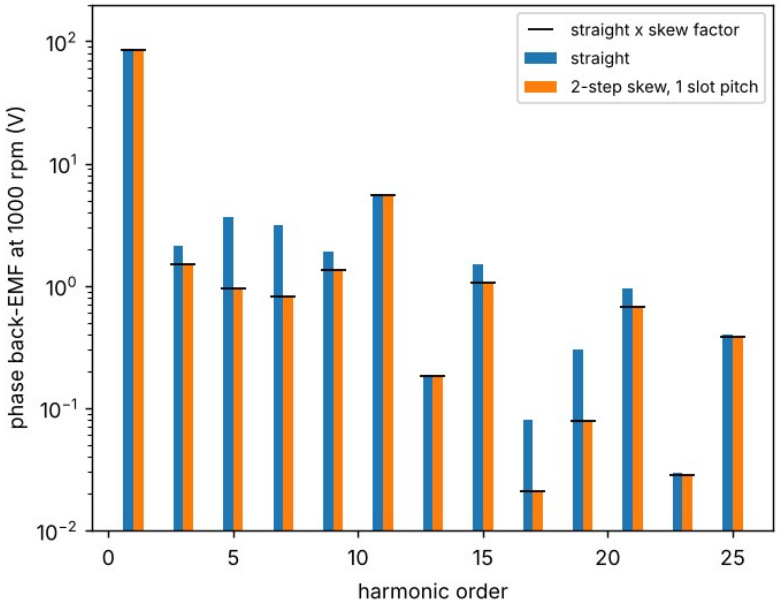
2 steps over 3.75 deg: cogging 0.62 N m, ripple 8.8 %, torque 338.8 N m.

3 steps over 5.625 deg: cogging 0.42 N m, ripple 7.9 %, torque 335.3 N m.

Back-EMF fundamental x0.96593 vs closed-form skew factor 0.96593; THD 9.3 -> 7.2 %.







Limits: multi-slice: no inter-segment axial flux, no end effects.

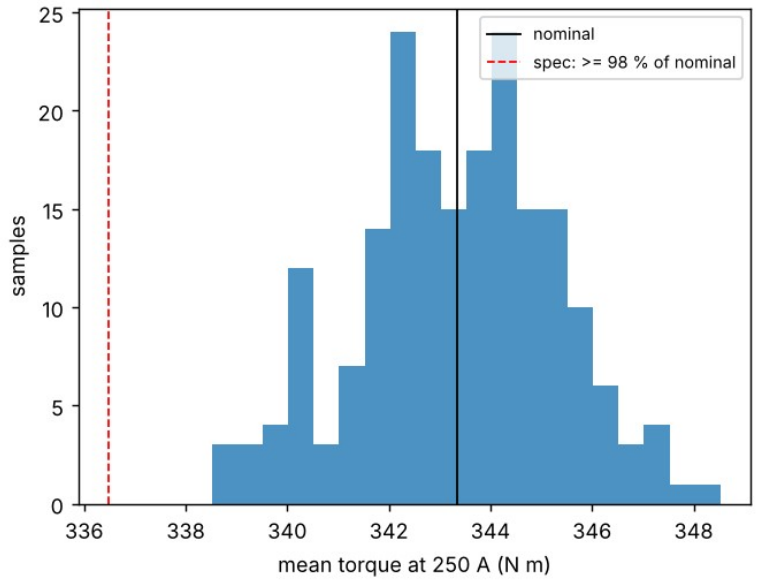
Manufacturing tolerances

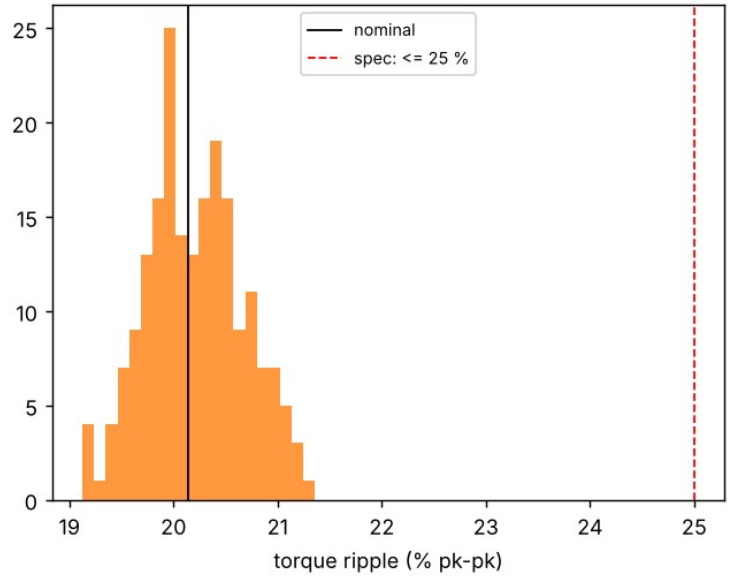
200 Monte-Carlo samples (JCGM 101), one nonlinear FEA each: remanence, magnet thickness and width, air gap.

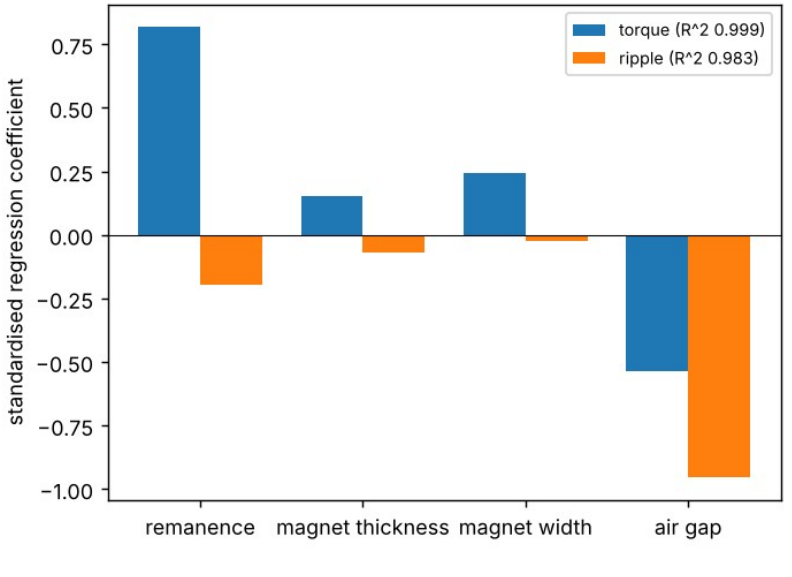
Torque 343.3 +- 1.96 N m; ripple 20.2 +- 0.47 %.

Largest effects: remanence on torque (SRC 0.82), air gap on ripple (SRC -0.95); linear fits R^2 0.999 / 0.983.

Yield against the stated specification 1.000 (Wilson 95 %: 0.981 to 1.000).







200 Monte-Carlo samples (JCGM 101), one FEA each; normal inputs, sigma: remanence 1 %, thickness 0.025 mm, width 0.05 mm, air gap 0.025 mm (stated assumptions, not measured tolerances). Joint probability of meeting both specifications 1.000 (Wilson 95 %: 0.981-1.000).

Limits: the tolerance sigmas are assumptions, not measured spreads.

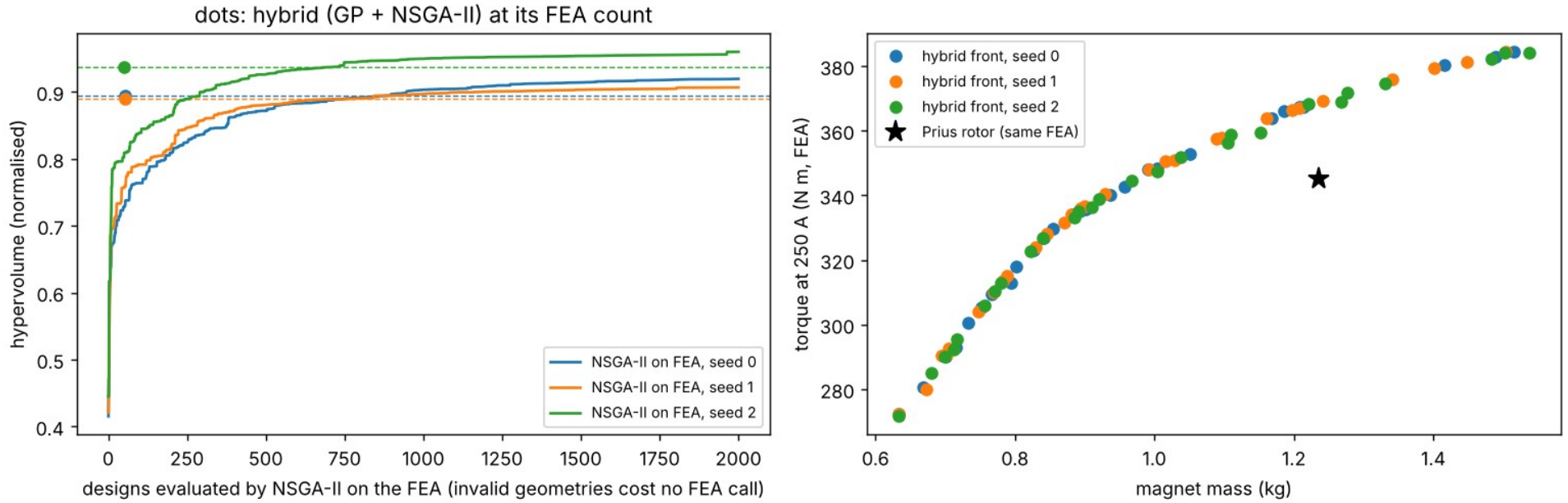
Rotor design: FEA + Gaussian process + NSGA-II

Five rotor variables, maximise torque, minimise magnet mass, ripple <= 30 %; every front design re-evaluated by a finer FEA.

FEA calls to equal hypervolume, hybrid vs NSGA-II on the FEA: seed 0: 53 vs 711 (92.5 % saved); seed 1: 52 vs 601 (91.3 % saved); seed 2: 51 vs 605 (91.6 % saved).

Finer FEA: all front designs feasible, torque within 1.1 %, ripple up to 30 % higher (6 angles under-sample it).

Designs beating the Prius rotor sit at the 1.0 mm bridge bound.



Hypervolume of the feasible FEA-evaluated designs (objectives normalised by the ideal and nadir of both runs of a seed). FEA-call saving of the hybrid at equal hypervolume: 92.5 %, 91.3 %, 91.6 %. Right: coarse FEA (288 gap nodes, 6 angles) of the hybrid fronts; every design re-evaluated at 480 nodes, 12 angles (torque within 1.1 %, ripple up to 30 % higher). The fronts sit at the 1.0 mm bridge bound: the mechanical strength of the bridge is not a constraint here.

Limits: no bridge stress, demagnetisation or speed constraints: shows the method and the FEA-call saving, not a better rotor.

Exchange formats and limits

In: pyleecan machine files (the real Toyota_Prius.json reads to exactly the model). Out: pyleecan JSON, FEMM Lua, Elmer case, gmsh/GetDP, MATLAB .mat, DXF, VTK, CSV/Parquet/HDF5 data contracts.

What is verified: closed forms (FEM, filaments, thermal, skew factor, drive), a second FEA code, a published torque curve.

Not verified: any measurement of this motor; FEMM/pyleecan/Elmer runs; cogging to better than about 25 %; thermal data; magnet grade.

Next: test-bench data (torque, EMF, temperatures) to calibrate thermal and loss models; a 3-D coupled end-region solution at load; bridge stress as a design constraint.

Limits: see the Limits section of docs/guides/electromagnetics.md.